

MASTER IN COMPUTER SCIENCE

Team 2: A Living Archive

From handwritten pages to a searchable map, with every result traced back to the original page.

5 Contributors

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Old handwriting is hard to search

Historians read manuscripts page by page. Our app lets them search the text, see where things happened, and always check the original.

1 · READ

Handwriting to text

Each page image becomes lines of text.

2 · FIND

Places in the text

Old place names become real places with a country.

3 · SHOW

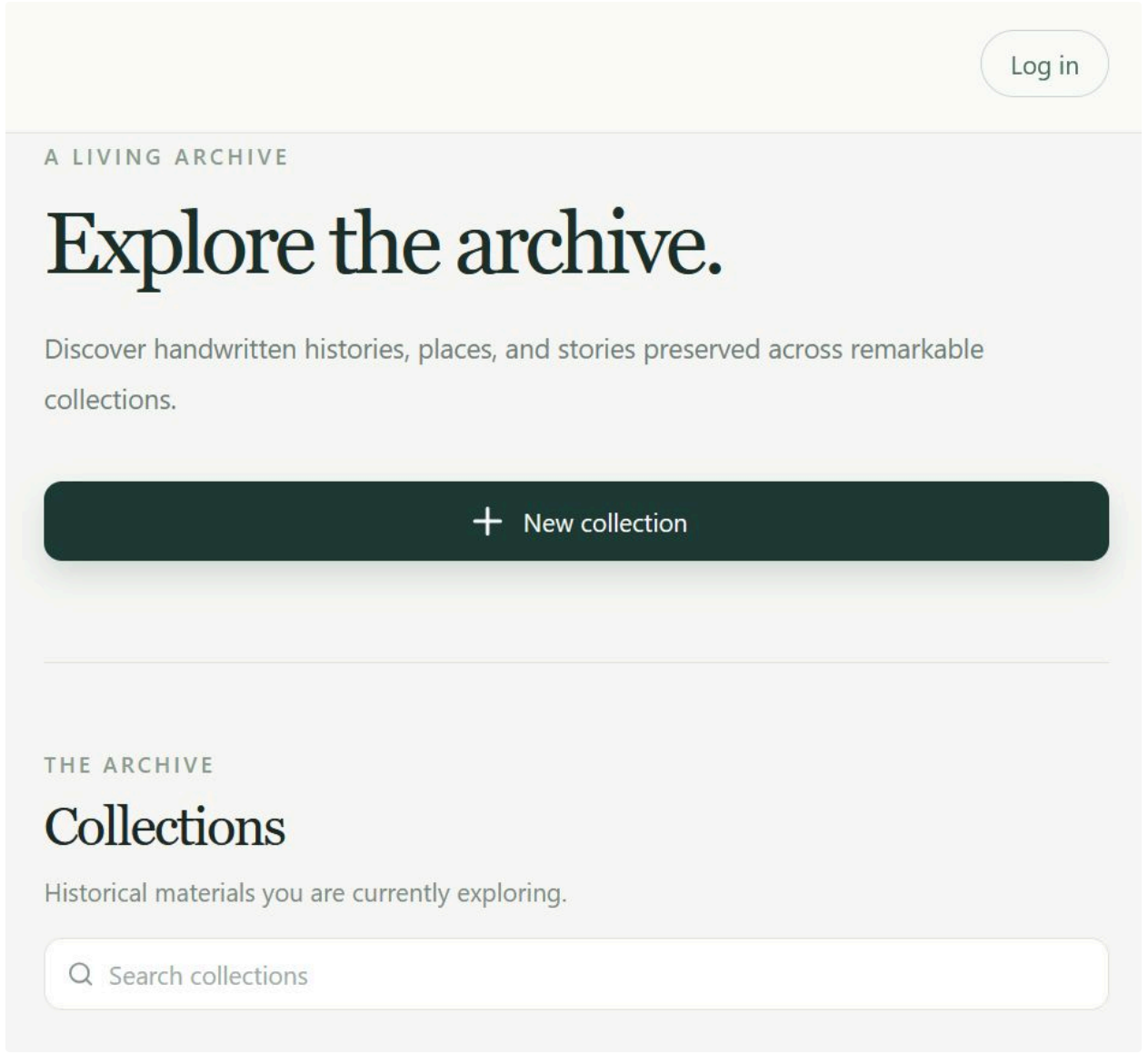
Map and evidence

Click a country, then open the original page.

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Demo

https://manuscript.ethanganz.com/



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Technology Stack

Database: PostgreSQL using Supabase

Backend: Python

Frontend: JavaScript



Image Sources: PostgreSQL - https://wiki.postgresql.org/wiki/Logo

Python - www.python.org/community/logos/

JavaScript - https://commons.wikimedia.org/wiki/File:Unofficial_JavaScript_logo_2.svg

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User Stories

- From a Backlog of Ideas
- Organized on ClickUp
- With Dependencies
- Different Priorities and Topics

Stories are self-assigned
by team members each sprint

ClickUp

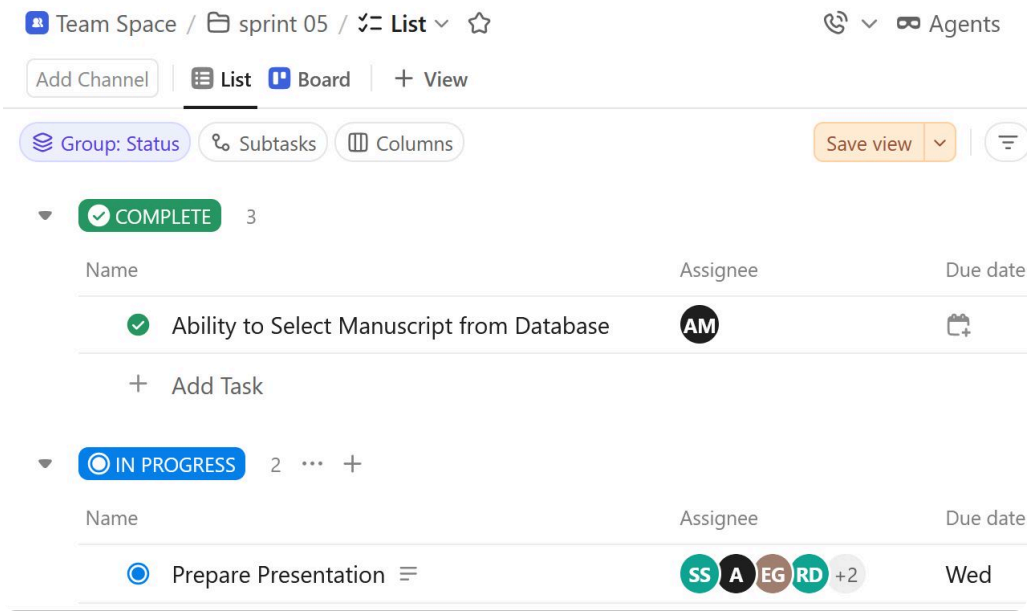


Image Source: https://www.prnewswire.com/news-releases/clickup-launches-the-https://app.clickup.com/

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User Stories

Example 1:

"As a user, I want to upload a manuscript."

Topic: General - File Management

Feature: Upload a file

Depends On: Website, Backend

Blocks: Almost Everything

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User Stories

Example 1:

"As a user, I want to upload a manuscript."

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Example 2:

"As a user, I want to see which locations are mentioned in a manuscript."

Topic: Locations

Features: Display Location References

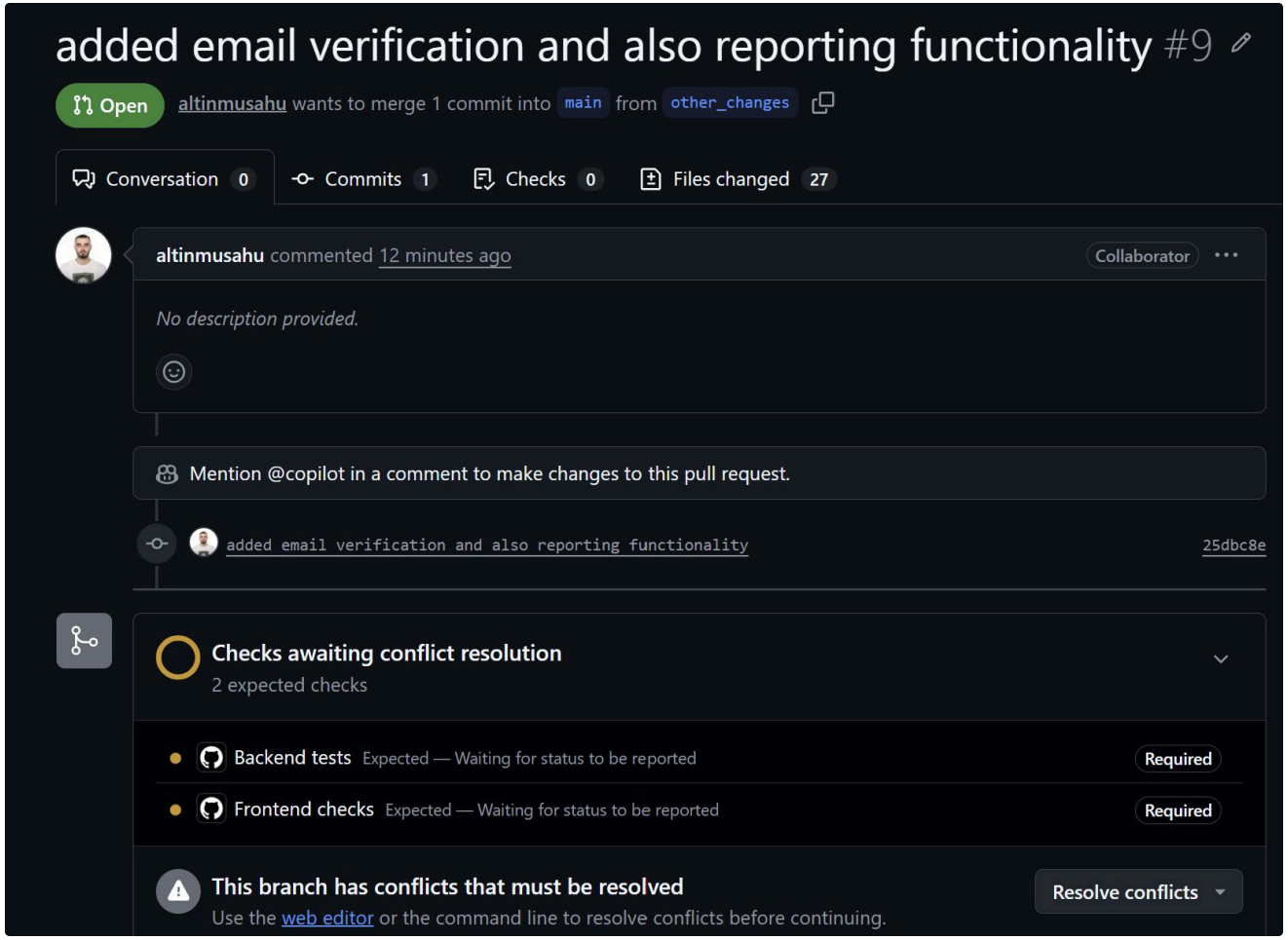
Depends On: Digital Text

Blocks: Show Locations on the Map

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Continuous Integration

GitHub Pull Requests to main



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GitHub Actions

build
succeeded 2 minutes ago in 5m 20s

- Set up job
- Checkout repository
- Set up QEMU
- Set up Docker Buildx
- Log in to GitHub Container Registry
- Read backend image version
- Read frontend image version
- Build and push backend image
- Build and push frontend image
- Trigger n8n deploy webhook
- Post Build and push frontend image
- Post Build and push backend image
- Post Log in to GitHub Container Registry
- Post Set up Docker Buildx
- Post Set up QEMU
- Post Checkout repository
- Complete job

GitHub Registry

Recent tagged image versions

latest 95594316def2c447034edc022e92fc634d5851c9 0.0.4
Published 5 minutes ago · Digest ...

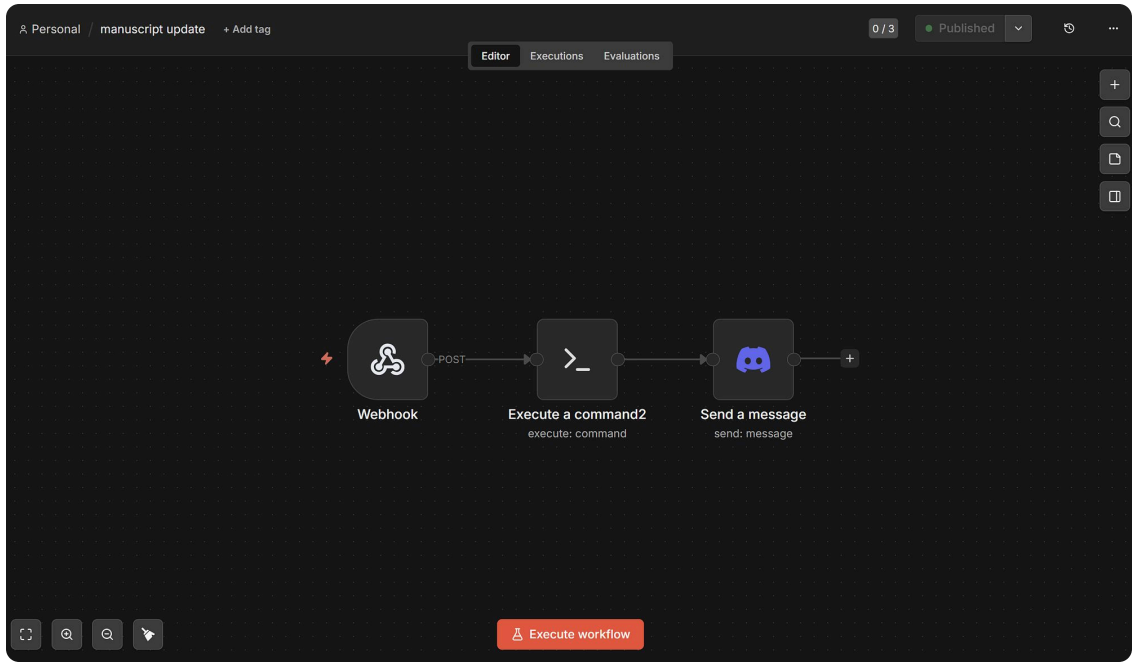
74547c4cc19e733560511ce08e6cd309aed196aa
Published 33 minutes ago · Digest ...

04e87dcd426a2766ba48b70a580ceb758da63ec7 0.0.3
Published about 1 hour ago · Digest ...

4d54f4d175a0ecfc171208919130fd83f04aff38
Published about 2 hours ago · Digest ...

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n8n Pipeline



Cloudflare Tunnels



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ML Workflow

1. TrOCR

Fine-tuned TrOCR to read German and Latin manuscript line images.

2. Place extraction

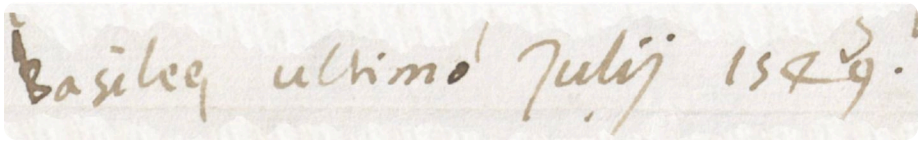
NER models detect place mentions in the text.

3. Geographic linking - validation need

Match historical names to geographic records and flag uncertain matches.

4. Map display - application goal

Show each linked place with its source evidence.



Real local inference example: "Basileę ultimo iulii 1544."

Extracted mention: Basileę

We implemented and evaluated transcription and extraction. Geographic links still need validation.

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Evaluation & Challenges

200 human-verified lines

- German and Latin manuscript lines, human-verified
- 50 calibration lines
- 150 test lines
- 9 reference place mentions in the test set

Challenges

- **Transcription errors affect place extraction.** In our small test set, TrOCR reproduced 4 of the 9 place mentions exactly. Errors in the other mentions can make place extraction more difficult.
- **False positives remain.** Some extracted words do not refer to places.
- **More evaluation is needed.** 9 place mentions are too few for strong model comparisons.

Later evaluations reused this test set and are exploratory.

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Outlook: Next Steps

Better recognition

- Improve TrOCR on difficult handwriting and historical place names
- Reduce false and missed mentions
- Expand evaluation using new verified data

Reliable mapping

- Resolve historical name variants and ambiguous matches
- Validate the final geographic links

User review and scale

- Let users inspect manuscript evidence and correct text or locations

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Thank you for your attention!

Questions & Discussion

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